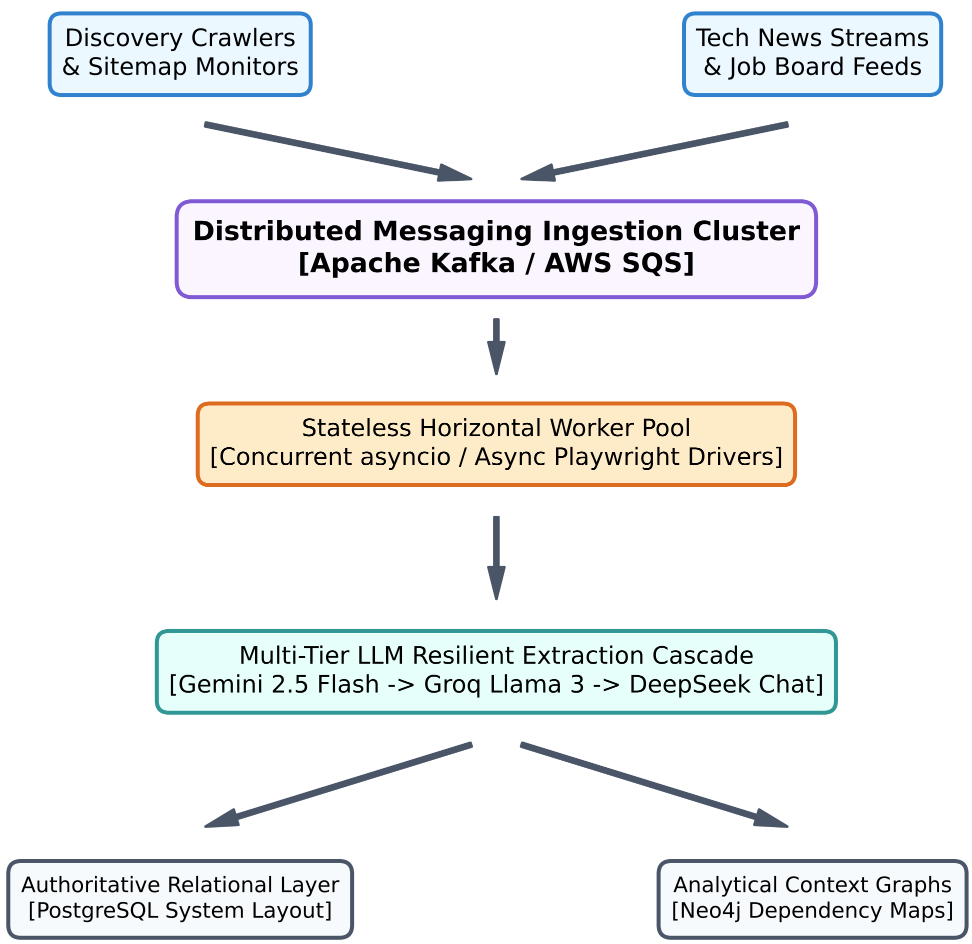
FrontierAtlas Ingestion Pipeline Architecture

Technical Production Design Document — Phase VI

1. Visual Infrastructure Flowchart Graph Map



Pipeline Scaling & Resiliency Blueprint Playbook

2. Horizontal Scaling Strategy (Goal: 500,000+ Records)

- Decoupled Architecture: Ingestion feeds are cleanly decoupled from processing tasks using an event-driven messaging topology via Apache Kafka topics partitioned by a hash of the source domain.
- Stateless Processing: Containerized consumer nodes scale horizontally using Kubernetes HPA profiles, avoiding vertical hardware performance degradation barriers during ingestion spikes.

3. Mitigation Strategy for Payload (413) & Rate Limits (429)

- Context Window Overflows (HTTP 413): Text streams go through token-aware chunking routines capped strictly at 80% of a model's limits, reserving the remaining space for schemas and self-correction.
- Thundering Herds (HTTP 429): Intercepted errors invoke an exponential backoff with full randomized jitter parameters: Delay = min(Max_Delay, Base_Delay * 2^attempt) + Uniform_Jitter.
- Multi-Tier Cascades: Automated error failovers jump from Gemini Flash directly to Groq or DeepSeek.

4. Distributed Freshness & Duplicate Elimination

- Probabilistic Filtering (Tier 1): Active lookups check incoming URLs against distributed Redis Bloom Filters instantly, dropping duplicate assets with less than 2ms network latency.
- Authoritative Locking (Tier 2): New targets are verified against a deterministic Redis Set equipped with a strict 48-hour sliding Time-to-Live (TTL) matrix to eliminate processing deadlocks.

5. Production Storage Framework Topology

- Transactional Relational Database (PostgreSQL): Securely parses structural Pydantic tables into indexes using highly optimized transactional native JSONB column query layouts.
- Analytical Graph Network (Neo4j): Stores real relationships across ecosystem primitives seamlessly (e.g., [Founder] -> CO_FOUNDED -> [Startup] -> LAUNCHED -> [Product]).